

Omri Azencot

✉ azencot@gmail.com
github.com/azencot-group
US Permanent Resident (Green Card)

Professional Summary

Senior Applied Scientist with 10+ years of experience developing large-scale machine learning systems, with a focus on generative AI, large language models, and multimodal data. Proven track record of designing end-to-end ML systems, including data pipelines, model training, evaluation, and inference, with emphasis on reliability and real-world performance. Experience developing systems that operate on complex, structured, and high-dimensional data, including sequential, multimodal, and document-like inputs. Strong background in experimentation, evaluation, and model validation, with a focus on building systems that improve through iterative feedback and data-driven refinement. Author of 40+ publications in top-tier venues (NeurIPS, ICML, ICLR).

Machine Learning Expertise & Impact

- Develop ML and LLM-based systems for large-scale, high-dimensional, and structured data, including sequential and multimodal signals.
- Design end-to-end ML pipelines, from data preprocessing and supervision design to model training, evaluation, and production-oriented inference.
- Build evaluation and validation frameworks, including offline testing, ablation studies, and regression detection for model quality and reliability.
- Develop systems that learn from user feedback and iterative corrections, improving performance over time.
- Apply modeling techniques to problems analogous to document understanding, decision support, and user-level prediction.
- Collaborate with engineering and research teams (Google, NVIDIA, Bosch AI) to develop scalable ML systems.

Core Competencies

Machine Learning	Deep Learning; Large Language Models (LLMs); Generative AI; Sequential Modeling; Multimodal Learning; Diffusion Models; Representation Learning
Systems	End-to-End ML Pipelines; Data Processing; Large-Scale Training; Model Evaluation & Validation; Experimentation; Efficient Inference; Scalable ML Systems
Applied Research	Experiment Design; Ablation Studies; Model Analysis; Algorithm Development; Data-Centric Modeling

Professional Experience

2020–Present **Principal Investigator & Senior Scientist, Ben-Gurion University of the Negev.**

- Develop ML and LLM-based systems for large-scale sequential and multimodal data, focusing on prediction, generation, and structured reasoning tasks.
- Designed and implemented end-to-end ML pipelines:
 - Transform large-scale time series, structured datasets, and text into model-ready representations
 - Generate semantic annotations using large-scale data processing and language models
 - Construct structured supervision and instruction-style datasets for training and evaluation

- Built evaluation and benchmarking frameworks for generative and LLM-based systems, including automated testing pipelines and regression detection.
- Developed systems incorporating iterative supervision and feedback signals, improving model performance through structured correction loops.
- Improved generative modeling performance by up to 58.17% on short-range tasks and 132.61% on long-range benchmarks compared to prior diffusion-based approaches.
- Developed unified models for data-scarce settings, achieving over 55% improvement in discriminative and FID-based metrics while using only 5% of the data.
- Reduced diffusion model inference from multi-step iterative sampling to single-step generation via distillation, significantly improving efficiency.
- Applied modeling approaches to problems analogous to document understanding, ranking, and decision-making under uncertainty.
- Collaborated with engineering and research partners (Google, Bosch AI, NVIDIA) to develop scalable ML systems.
- Led and mentored a team of researchers across multiple ML system development efforts.
- Published 40+ papers in top-tier venues.

2025–Present **Research Affiliate, ICSI Berkeley.**

Research on large-scale machine learning systems for multimodal data and generative modeling, with focus on LLM-based and generative approaches.

2017–2020 **Assistant Adjunct Professor, UCLA.**

Developed scalable machine learning algorithms for structured and sequential data, including operator-based and generative modeling methods.

Selected Technical Contributions

- NeurIPS 2025 Unified generative modeling for time series under data scarcity, improving performance across multiple benchmarks.
- NeurIPS 2025 One-step distillation of diffusion models, enabling single-step generation and efficient inference.
- ICLR 2026 Disentangled multimodal representations enabling controllable generation and improved generalization.
- NeurIPS 2024 Scalable diffusion-based modeling for time series generation with improved data efficiency.

Education

2011–2017 **Ph.D. in Computer Science, Technion.**

2005–2010 **B.Sc. in Computer Science and Mathematics, Technion.**